

EE 211: Analog Circuit Design

Course Code	EE 211
Course Name	Analog Circuit Design
L-T-P-C	2-0-2-3
Prerequisites	Network theory (EE203), Applied Electronics (IC 161)
Intended for	B. Tech (EE)
Distribution	Core for B. Tech (EE) & Elective for other UG programs.
Approval	32nd BoA (Dec 2019); 24th Senate (Feb 2020)
Recorded to	VL 211 by the 64th BoA (May 2026)
Discontinued	64th BoA (May 2026)

Course Contents

- **BJT/MOS single stage amplifiers, cascade and cascodes** [6 hrs]
 - Large signal and small-signal model, biasing, input and output impedance, operating point calculations and design, single ended BJT/CMOS amplifiers, cascade and cascode amplifiers
- **BJT/MOS Current mirrors** [3 hrs]
 - PVT independent normal current mirror, cascade current mirror, regulated current mirror and Wilson current mirror
- **Differential Amplifiers** [3 hrs]
 - MOS/BJT Differential Pair, qualitative large/small signal analyses, differential pairs with active loads and common-mode rejection
- **BJT/MOS Frequency Response**
 - High frequency model of single ended and differential amplifiers, Frequency response: magnitude and phase plot calculations [3 hrs]
- **Feedback theory** [3 hrs]
 - Properties of negative and positive feedback, loop gain calculations, types of amplifiers, voltage controlled current source (VCCS), current controlled current source (CCCS), current controlled voltage source (CCVS), voltage current voltage source (VCVS), stability analyses under negative feedback topology.
- **Output stages and power amplifiers** [3 hrs]
 - Emitter follower as power amplifier, push-pull amplifier, cross-over distortions, large signal considerations, heat dissipation, efficiency, Classes of amplifiers
- **Oscillators and phase locked loop** [4 hrs]
 - Barkhausen criteria of oscillation, bistable, monostable and astable multi vibrators, LC, relaxation, phase shift and Colpitts oscillators, phase locked loop concept and its understanding, signal generation and wave shaping circuits.
- **Data converters** [3 hrs]
 - Digital-to-analog converters (DAC): R-2R, current scaling and voltage scaling
 - Analog-to-digital converter (ADC): flash, SAR, single slope, dual slope, pipeline and sigma-delta modulator

Experiment modules

1. Introduction to laboratory: DSO and its advance features, XY-mode (Lissajous pattern), LTspice and required software.
2. Understanding of I_V characteristics of BJT/MOS transistor using net-listing in SPICE simulations.
3. Common emitter amplifier design.
4. Understanding of MOS class-AB push-pull amplifier using CD4007 IC.
5. Oscillator: design of ring, LC and phase shift oscillators.
6. Operation of PLL understanding using 565 IC.
7. Operation of SAR understanding using 0808/0809 I C.
8. Project on analog circuit application.

Text Books

1. Behzad Razavi, **Fundamentals of microelectronics**, Wiley, 2013.

References

1. A.S. Sedra and K.C. Smith, **Microelectronic Circuits-Theory & Applications**, 7th Edition, Oxford University Press, 2017.